

Concrete — 20 × 16 ft

Outdoor & Structural · Challenging · A full weekend with help

YOU NEED APPROXIMATELY

4.35 yd³

Recommended purchase: 4.50 yd³ including 10% waste

Project summary

Project area	320 sq ft
Slab thickness	4 in
Calculated volume	3.95 yd ³
Waste allowance	10%
Recommended purchase	4.50 yd ³

Materials & budget

ITEM	QUANTITY	UNIT PRICE	COST
Ready-mix concrete (delivered) Order quantity rounded up to the nearest supplier increment.	4.50 yd ³	\$165 per yd ³	\$743
Compactable base gravel Includes 10% compaction allowance.	4.50 yd ³	\$55.00 per yd ³	\$248
Welded wire mesh ESTIMATE Coverage allowance including overlap. Reinforcement needs vary by use and local practice.	336 sq ft	—	—
Form boards (2×4 or 2×6) ESTIMATE Perimeter plus 10% for corners and overlap.	79 linear ft	—	—
Form stakes ESTIMATE Roughly one every 3 ft, plus corners.	28 stakes	—	—
Estimated material total			\$990

Costs use ProjectKit planning prices or the prices entered by the project owner. They are not live retail prices.

Options compared

Ready-mix delivery

One truck, one pour. Best once you pass about a cubic yard.

Order quantity	4.50 yd ³
Estimated concrete cost	\$743

Bagged concrete

No delivery minimum, but mixing is the whole afternoon.

Bags needed	261 bags
Estimated concrete cost	\$1,697

Shopping list

- ☐ Ready-mix concrete (delivered)
4.50 yd³
- ☐ Compactable base gravel
4.50 yd³
- ☐ Welded wire mesh
336 sq ft
- ☐ Form boards (2×4 or 2×6)
79 linear ft
- ☐ Form stakes
28 stakes
- ☐ Screed board (straight 2×4 longer than the slab width)
- ☐ Expansion joint material where the slab meets existing concrete
- ☐ Bull float or magnesium float
- ☐ Concrete edger and groover
- ☐ Curing compound or plastic sheeting
- ☐ Waterproof gloves, eye protection, knee pads
- ☐ Rebar chairs or mesh supports

Project sequence

1. Mark out the slab and check it for square by measuring both diagonals.
2. Call for utility locates before you dig.
3. Excavate to slab thickness plus base depth.
4. Add and compact the gravel base in layers.
5. Build and brace the forms, setting the slope for drainage.

- 6. Place reinforcement on chairs so it sits mid-slab.
- 7. Re-check dimensions, depth, and slope before the concrete arrives.
- 8. Place the concrete and screed it level with the forms.
- 9. Float, edge, and finish once the bleed water is gone.
- 10. Cure it — keep it damp or covered for several days.
- 11. Strip the forms and backfill the edges.

Assumptions used

Waste allowance	10%
Concrete price	\$165 per yd³
Bag yield	0.45 cu ft per bag
Reinforcement	Welded wire mesh
Base course	4 in compacted gravel

How it was calculated

Area	Length × Width
Exact	
Volume	Area × (Thickness ÷ 12)
Exact	
Cubic yards	Cubic feet ÷ 27
Exact	
Waste adjusted	Calculated quantity × (1 + Waste percentage)
Exact	
Order rounding	Rounded up to the next 0.25 yd³
Assumption	
Base gravel	Area × (Base depth ÷ 12) × 1.10 compaction allowance
Assumption	

Before you start

Concrete sets on its own schedule. Have every tool, helper, and form check done before the truck arrives.

Slab thickness, reinforcement, and base depth requirements vary by use and by local code. Verify before you pour.

Slab thickness, reinforcement, base depth, and footing requirements vary by use, soil, climate, and local building code. Verify structural and code requirements before you build.

Notes

ProjectKit provides planning estimates only. Actual requirements, costs, installation methods, structural requirements, permits, safety requirements, and building codes vary. Verify critical specifications before purchasing materials or beginning work.